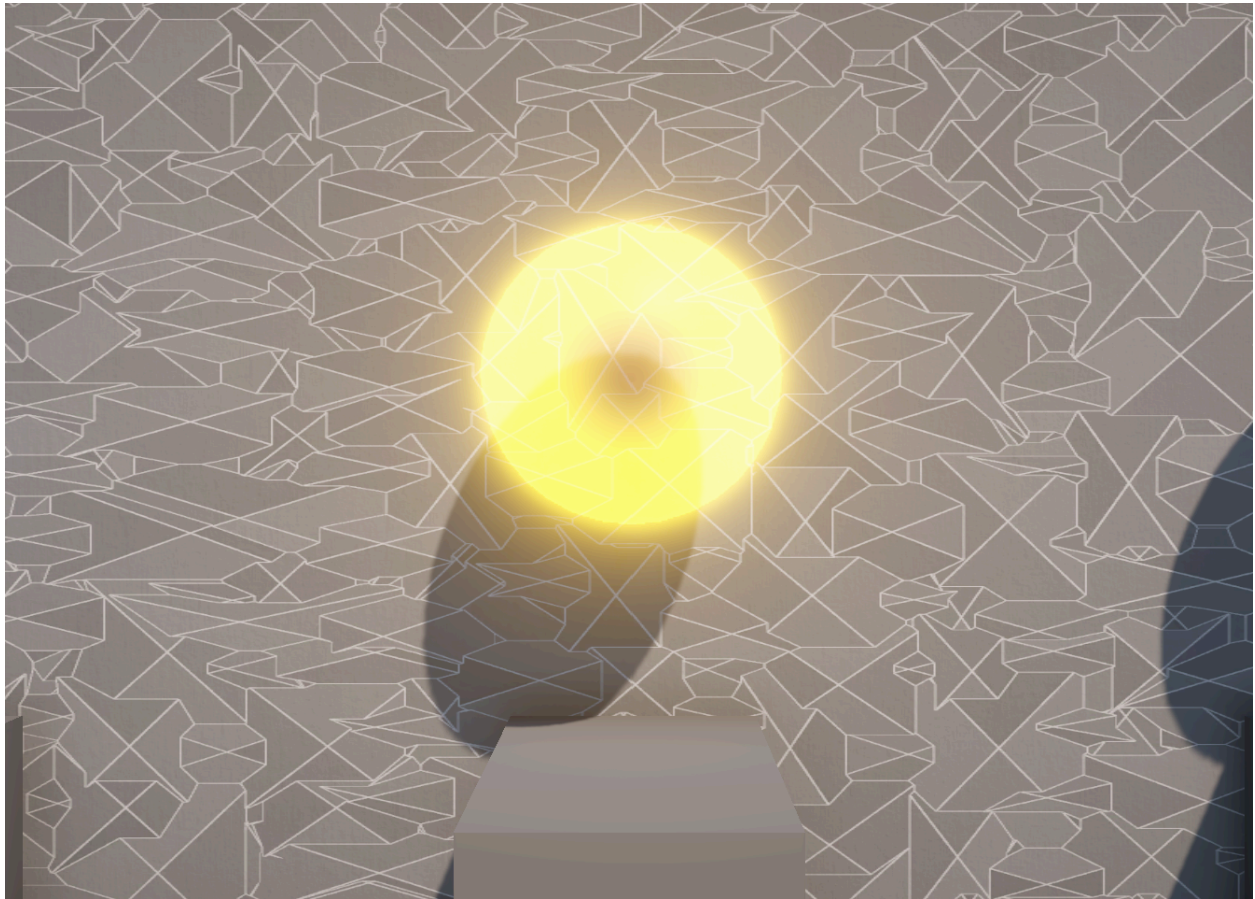
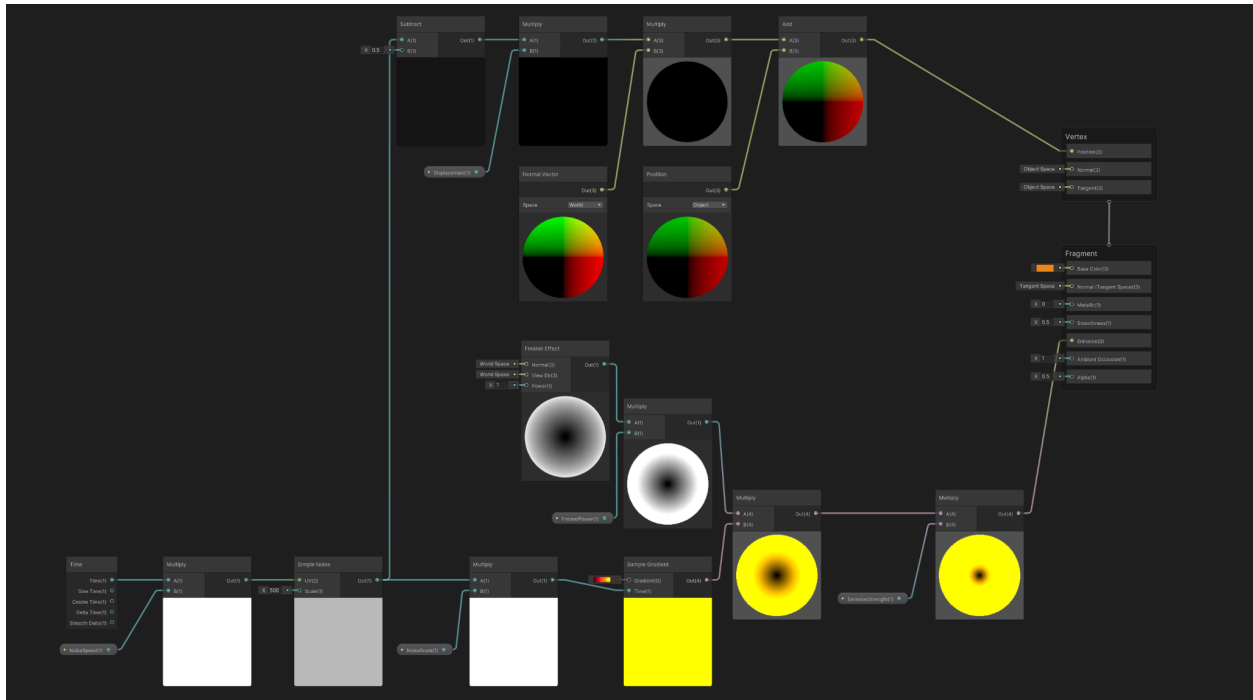


Fireball - Unity Shader Graph



This shader creates a bright ball of light that could be used as the core of a fireball.

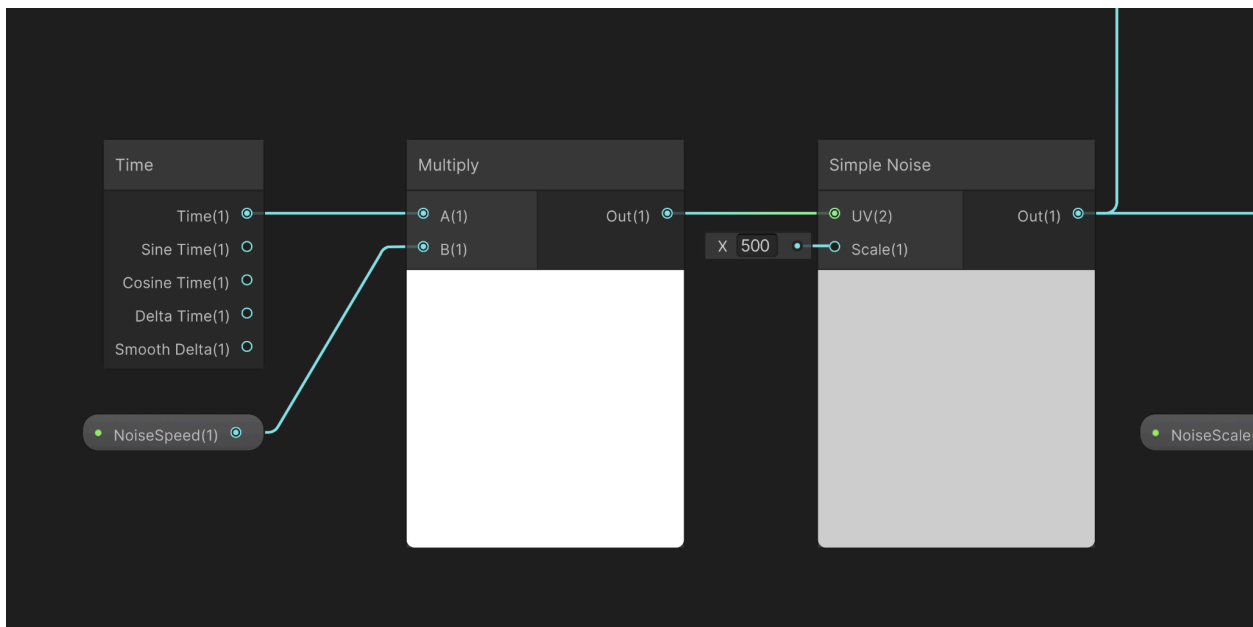
The layout of the shader is as follows:



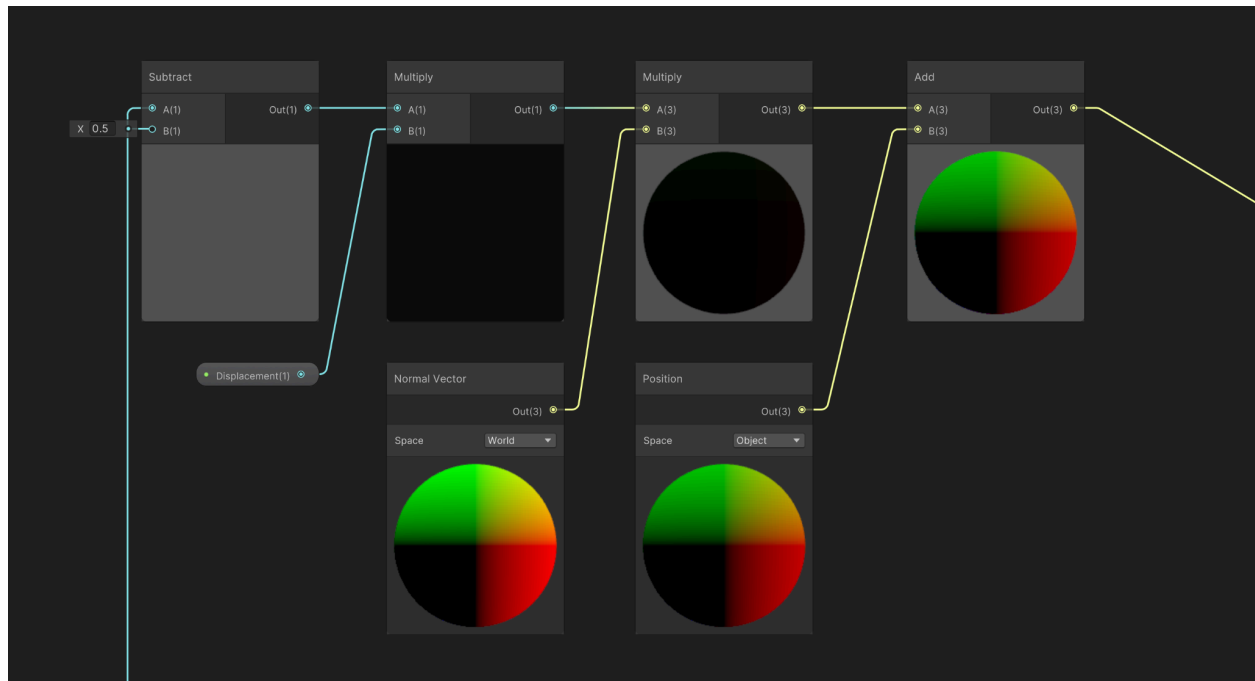
There are a few components to this shader:

- Size Distortion
- Flare
- Noise

Let's look at the noise first, which is the basis for the other two:

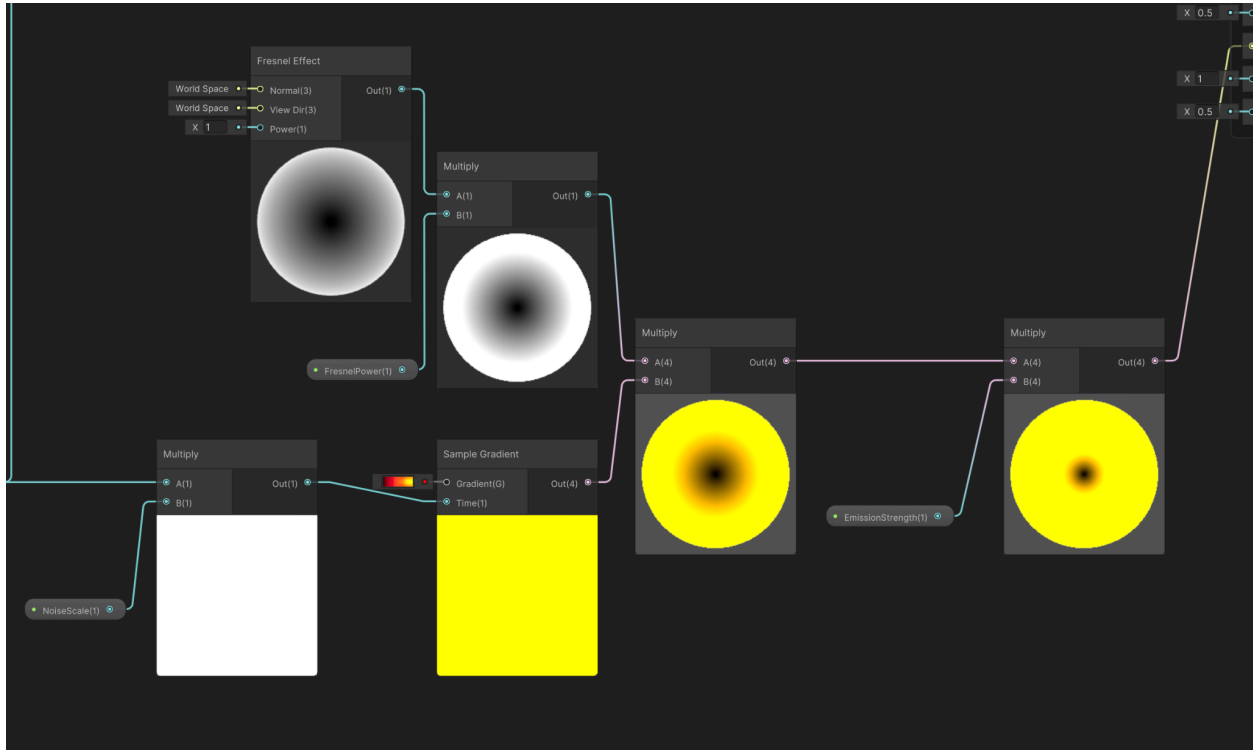


We take Time and multiply it with a float for speed to determine how fast our noise is. Then we pass that float into the UV for a Simple Noise node. This gives us the noise map we will use for our other 2 components.



Next is the size distortion. Take the noise output and pass it into a Subtract Node, subtracting 0.5. This effectively creates an inversion of the noise. Multiply that by a float for the magnitude of displacement. Then multiply that by a normal vector, and add it with an Object Space Position Node.

This effect will cause the ball to flicker in size, rapidly expanding and shrinking. Because this effect is quite strong, the Displacement should be a small value (~ 0.05).



Finally, we have the flare color. Take the Sample Noise and multiply it by a Scale (recommended: 2-5). Then feed that into the Time input of a Sample Gradient Node.

Create a Fresnel Effect Node. This will determine the shape of the output glow. Multiply it by a float to increase its magnitude. Then multiply the Fresnel Node with the Sample Gradient to give it color. You can choose to multiply it again to increase the Emission Strength (Recommended: 3-8).

Plug the final Fresnel Node into the Emission of the Shader. Then plug the Size Distortion into the Vertex Position.